Chapter 18, Problem 13P

Problem

Find the Fourier transform of the following signals:

(a) $f(t) = \cos(at - \pi/3)$, $-\infty<t<\infty$

(b) $g(t) = u(t + 1) \sin \pi t$, $-\infty<t<\infty$

(c) $h(t) = (1 + A \sin at) \cos bt$, $-\infty<t<\infty$, where A, a, and b are constants

(d) $i(t) = 1 - t$, $0<t<4$

Step-by-step solution

Step 1 of 3

(A) we know that $\mathrm{F}[\omega sat] = \pi[\delta(\omega-a)+\delta(\omega+a)]$

using the time shifting property,

$$\mathrm{F}\left[\cos a\left(t-\frac{\pi}{3a}\right)\right] = \pi e^{\frac{-j\omega\pi}{3a}}\left[\delta(\omega-a)+\delta(\omega+a)\right]$$

$$= \pi e^{\frac{-j\pi}{3a}}\delta(\omega-a)+\pi e^{\frac{j\pi}{3a}}\delta(\omega+a)$$

Step 2 of 3

(B) $\sin\pi(t+1) = \sin \pi t\cos\pi + \cos\pi t\sin\pi$

$= -\sin \pi t$

$g(t) = -u(t+1)\sin(t+1)$

$Let \quad X(t) = u(t)\sin t, then X(\omega) = \dfrac{1}{(j\omega)^2+1} = \dfrac{1}{1-\omega^2}$

Using the time slifting property,

$$\mathrm{G}(\omega) = \frac{-1}{1-\omega^2}e^{j\omega} = \frac{e^{j\omega}}{\omega^2-1}$$

Step 3 of 3

(C) $Let \;\; y(t) = 1 + A\sin at\;, \quad then \;\; Y(\omega) = 2\pi\delta(\omega) + j\pi A[\delta(\omega+a)]$

$h(t) = y(t)\cos bt$

using the Modulation property

$$\mathrm{H}(\omega) = \frac{1}{2}\left(Y[\omega+b]+Y[\omega-b]\right)$$

$$H(\omega) = \pi[\delta(\omega+b)+\delta(\omega-b)] + \frac{j\pi A}{2}\begin{bmatrix}\delta(\omega+a+b)-\delta(\omega-a+b)+\delta(\omega+a-b)\\ -\delta(\omega-a-b)\end{bmatrix}$$

$$I(\omega) = \int_o^4 (1-t)e^{-j\omega t}\,dt$$

$$= \frac{e^{-j\omega t}}{-j\omega} - \frac{e^{-j\omega t}}{-\omega^2}(-j\omega t-1)\Bigg|_0^4$$

$$= \frac{1}{\omega^2} - \frac{e^{-4j\omega}}{j\omega} - \frac{e^{-j4\omega}}{\omega^2}(j4\omega+1)$$